CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

FUNDAMENTALS OF DATA SCIENCE

Case Study Report

Case Study 2: Clustering Analysis of Airline Customers

Case Study 5: Wine Quality Assessment Using Physicochemical Properties

Name	:	Matam Shekar
Roll No.	:	160124733186
Section	:	CSE-3
Year	:	III Year
Subject	:	Fundamentals of Data Science
Faculty Name	:	S. Durga Devi

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About This Report

This report presents two case studies from the subject Fundamentals of Data Science. Case Study 2 groups the customers of an airline into segments using three clustering methods. Case Study 5 studies how the laboratory measurements of wine are related to its quality score and builds a regression model to predict that score.

Both case studies were carried out in Python using one Jupyter Notebook. The main libraries used were pandas and NumPy for data handling, Matplotlib and Seaborn for graphs, SciPy for statistical tests, and scikit-learn for the clustering and regression models. All graphs in this report are drawn in black and grey. The notebook, the Python file and the data files are available in the GitHub repository submitted along with this report.

Case Study 2: Clustering Analysis of Airline Customers

1. Problem Statement

East-West Airlines wants to know more about its customers. It is especially interested in how often customers fly, how they earn and use frequent flyer rewards, and how they use the airline's credit cards. The task is to divide the customers into groups, called segments, so that each group can be understood and served better.

Three clustering algorithms were applied to the data: K-Means, Hierarchical clustering and DBSCAN. The clusters found by each method were studied, their characteristics were described, and their quality was measured with the silhouette score.

2. Methodology

2.1 Dataset

The dataset contains 3,999 customers and 12 columns. One column is the customer ID and the other 11 describe the customer. Table 1 explains these columns. The three credit card columns are not exact miles. They are bins from 1 to 5, where 1 means under 5,000 miles and 5 means over 50,000 miles earned in the past 12 months.

Table 1: Variables in the East-West Airlines dataset

Variable	Meaning
Balance	Miles eligible for award travel
Qual_miles	Miles counted towards Topflight status
cc1_miles	Miles earned with the frequent flyer credit card (bin 1 to 5)
cc2_miles	Miles earned with the Rewards credit card (bin 1 to 5)
cc3_miles	Miles earned with the Small Business credit card (bin 1 to 5)
Bonus_miles	Miles earned from non-flight bonus transactions in 12 months
Bonus_trans	Number of non-flight bonus transactions in 12 months
Flight_miles_12mo	Flight miles in the past 12 months
Flight_trans_12	Number of flight transactions in the past 12 months
Days_since_enroll	Days since the customer joined the programme
Award?	1 if the customer has received an award, otherwise 0

2.2 Data Preparation

- **ID column:** The ID is only a label and carries no customer behaviour, so it was removed.
- **Missing values and duplicates:** There were no missing values. One duplicate row was found and removed, which left 3,998 customers.
- **Outliers:** A few customers have extremely large values. For example, the largest Balance is 1,704,838 miles while the middle value is only 43,138. Such points pull the cluster centres towards themselves. Customers whose z-score was above 3 in any of Balance, Bonus_miles, Bonus_trans, Flight_miles_12mo or Flight_trans_12 were removed. This removed 249 customers (about 6.2%) and left 3,749 customers.
- **Scaling:** The variables are measured on very different scales. Without scaling, a large-valued column such as Balance would decide the clusters on its own. All columns were therefore standardised to have a mean of 0 and a standard deviation of 1.

2.3 Clustering Methods and Evaluation

- **K-Means:** Customers are placed into k groups around centre points. The value of k was chosen using the elbow curve and the silhouette score for k from 2 to 10.
- **Hierarchical clustering:** Customers are merged step by step into a tree. A dendrogram was drawn using Ward linkage, and the Ward, complete, average and single linkage methods were compared.

- **DBSCAN:** Groups are formed from dense regions of customers and sparse points are marked as noise. It needs two settings, eps (the neighbourhood distance) and minPts (the minimum number of points). A k-distance plot and a grid of different values were used to choose them.
- **Silhouette score:** This score lies between -1 and 1. A higher value means customers are close to their own cluster and far from other clusters. For DBSCAN, noise points were left out when the score was calculated.
- **Visualisation and interpretation:** Principal Component Analysis (PCA) was used to show the clusters on a two-dimensional plot. The clusters were described by the average value of each variable in the original units.

3. Statistical Findings

Table 2 gives a summary of the main numeric variables before outlier removal. For every variable except Days_since_enroll, the mean is much larger than the median and the skewness is positive. This means most customers have small values and a small number of customers have very large values.

Table 2: Summary of the main numeric variables (3,998 customers)

Variable	Mean	Median	Std. Dev.	Maximum	Skewness
Balance	73,619	43,138	100,782	1,704,838	5.00
Bonus_miles	17,149	7,183	24,152	263,685	2.84
Bonus_trans	12	12	10	86	1.16
Flight_miles_12mo	460	0	1,400	30,817	7.45
Flight_trans_12	1	0	4	53	5.49
Days_since_enroll	4,119	4,098	2,065	8,296	0.12

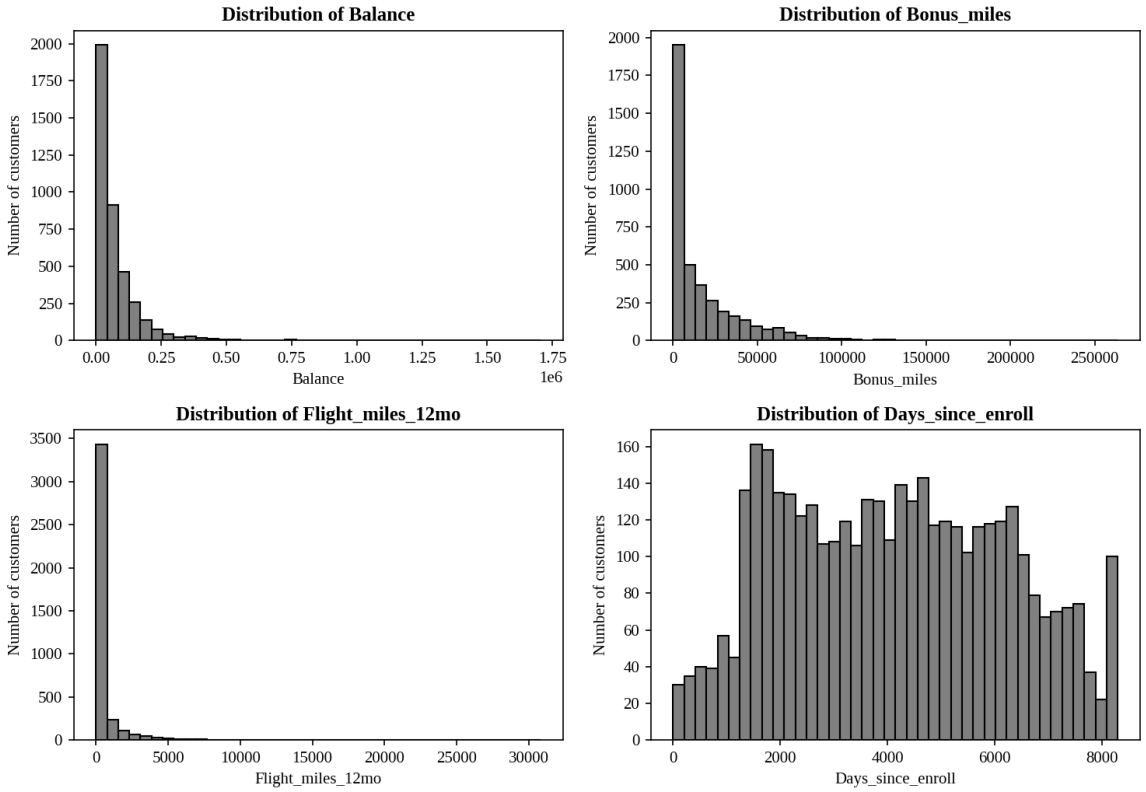


Figure 1: Distribution of Balance, Bonus miles, Flight miles and Days since enrolment

Interpretation: Balance, bonus miles and flight miles are strongly right-skewed, with skewness values of 5.0, 2.8 and 7.5. Most customers sit near zero and a long tail stretches to very high values. More than two thirds of customers (68%) flew no miles at all in the last 12 months. Days since enrolment is spread fairly evenly (skewness 0.12), which shows that customers joined over many years, from a few days up to about 8,300 days (nearly 23 years).

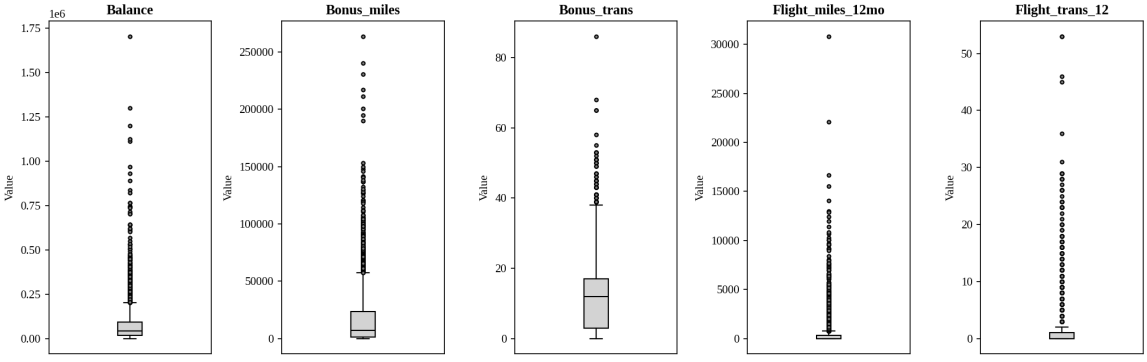


Figure 2: Box plots of five numeric variables before outlier removal

Interpretation: Every variable has a long line of dots above the box, which are extreme values. The boxes themselves are very small compared with the highest values. This confirms that outliers are present and supports the decision to remove them before clustering.

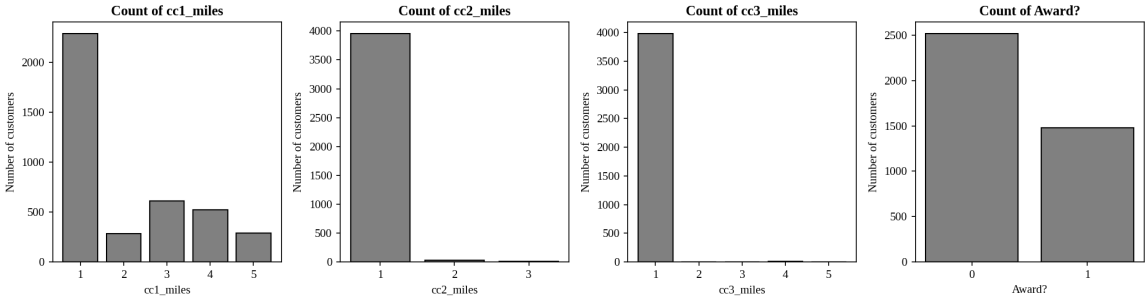


Figure 3: Number of customers in each credit card mile bin and award status

Interpretation: The frequent flyer card (cc1) is the only card that is widely used. About 57% of customers are in bin 1 (under 5,000 miles) and about 7% are in the top bin. In contrast, 98.9% of customers are in bin 1 for the Rewards card (cc2) and 99.5% for the Small Business card (cc3), so almost nobody earns miles with them. Regarding awards, 37% of customers have received an award and 63% have not.

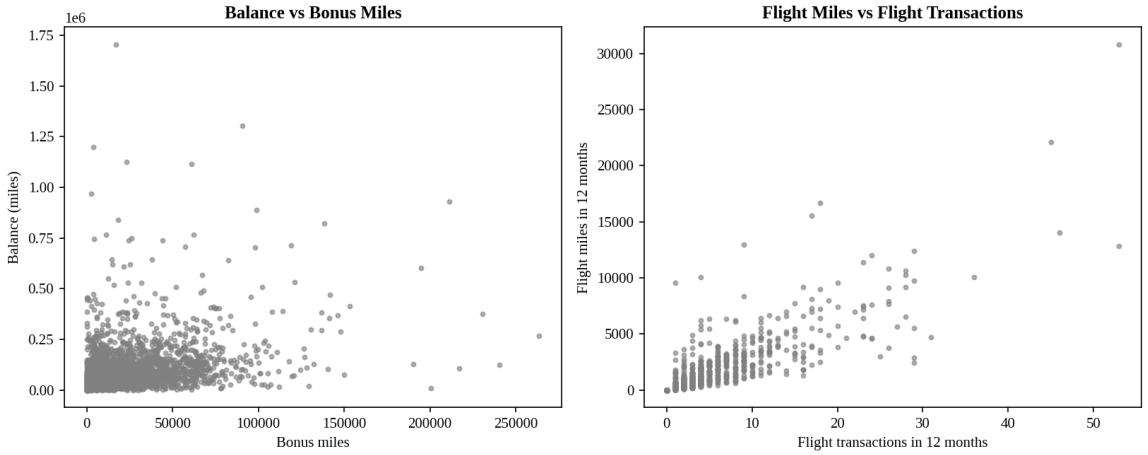


Figure 4: Balance against bonus miles, and flight miles against flight transactions

Interpretation: Balance generally goes up as bonus miles go up, but the points are widely scattered, so the link is only moderate. Flight miles and flight transactions rise together almost in a straight line, which shows that customers who fly more often also collect more flight miles.

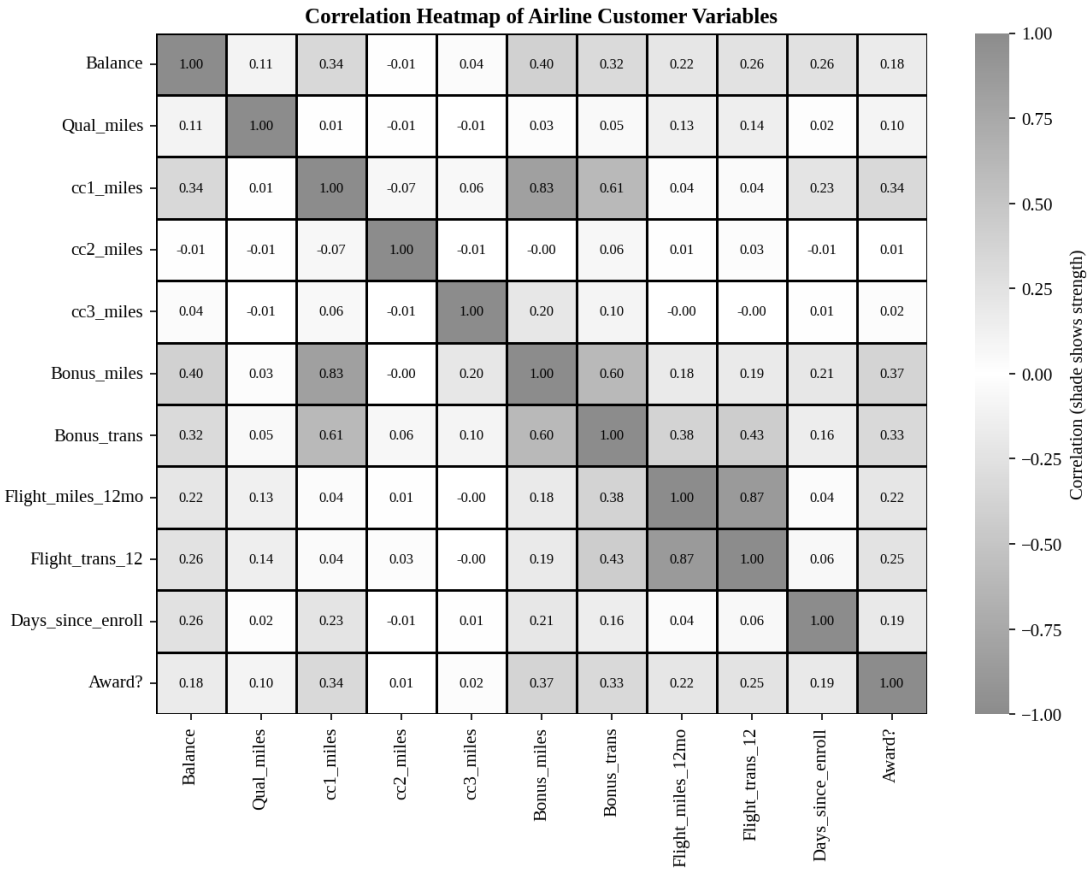


Figure 5: Correlation heatmap of the airline customer variables

Interpretation: The darker the box, the stronger the relationship, and the number shows whether it is positive or negative. The strongest relationships are between Flight_miles_12mo and Flight_trans_12 (0.87), between cc1_miles and Bonus_miles (0.83), and between Bonus_trans and both Bonus_miles and cc1_miles (about 0.60). The flying variables and the credit card and bonus variables are almost unrelated to each other (correlations between 0.04 and 0.19). This suggests there are different types of customers, those who mainly fly and those who mainly earn miles through cards, which is what clustering should uncover. The cc2 and cc3 columns have almost no correlation with anything because so few customers use those cards.

4. Model Evaluation

4.1 K-Means Clustering

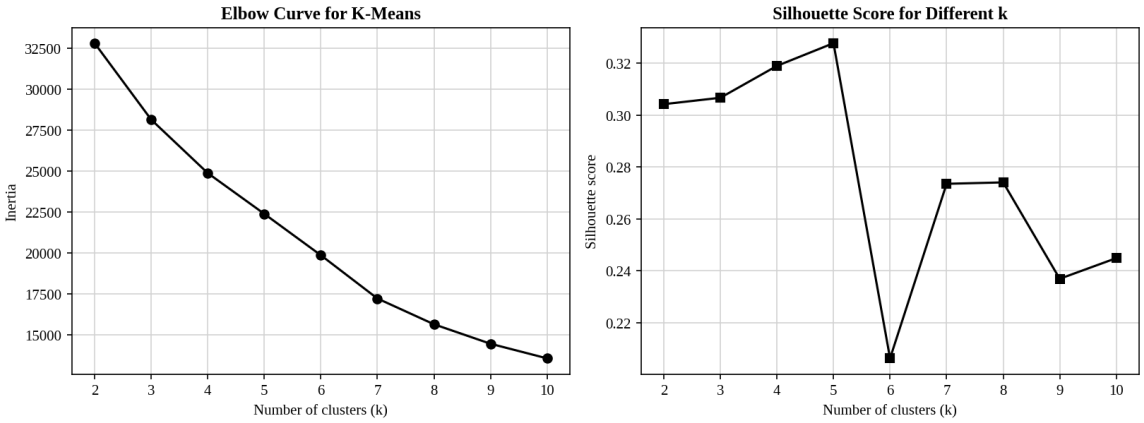


Figure 6: Elbow curve and silhouette score for K-Means with k from 2 to 10

Table 3: Inertia and silhouette score for different values of k

k	Inertia	Silhouette score
2	32,827	0.304
3	28,160	0.307
4	24,895	0.319
5	22,405	0.328
6	19,880	0.206
7	17,238	0.273
8	15,664	0.274
9	14,474	0.237
10	13,582	0.245

Interpretation: The inertia falls steadily as k grows and there is no sharp elbow, but the curve bends gently around k = 4 and k = 5. The silhouette score rises from 0.304 at k = 2 to 0.328 at k = 5 and then drops sharply to 0.206 at k = 6. The score at k = 4 (0.319) is very close to the best score, and four groups are easier to explain than five, so k = 4 was selected. The final K-Means model has a silhouette score of 0.319, with clusters of 2,229, 1,231, 251 and 38 customers.

4.2 Hierarchical Clustering

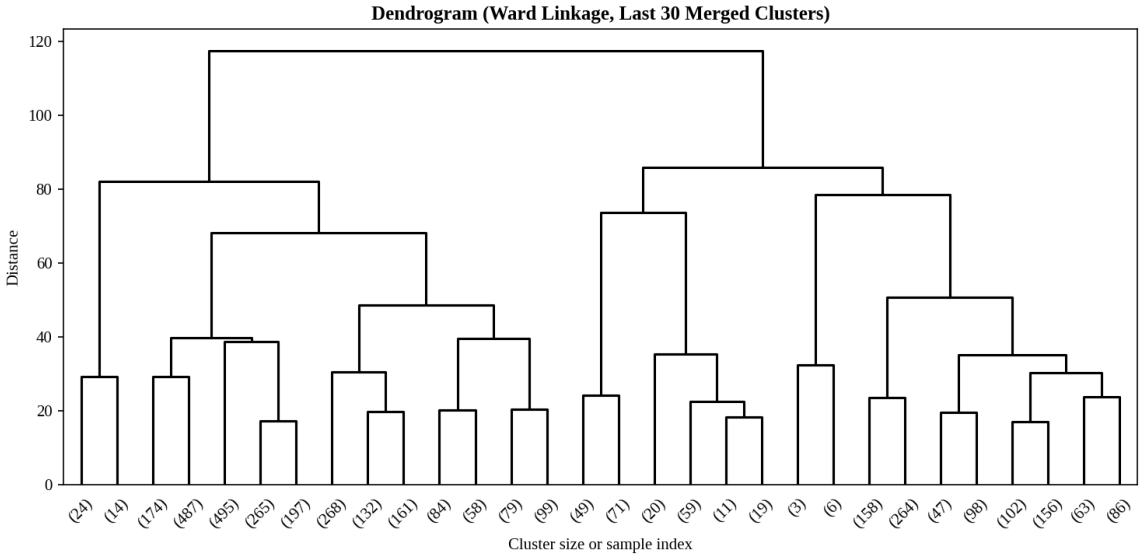


Figure 7: Dendrogram using Ward linkage (last 30 merged clusters)

Interpretation: The numbers in brackets show how many customers are inside each branch. The tallest vertical lines at the top show that the biggest differences appear when the data is divided into two to four groups, so four clusters were used to stay consistent with K-Means.

Table 4: Comparison of linkage methods

Linkage	Clusters	Silhouette	Largest cluster	Smallest cluster
Ward	3	0.271	2,537	229
Ward	4	0.287	2,499	38
Ward	5	0.282	2,499	9
Complete	3	0.726	3,740	3
Complete	4	0.631	3,702	3
Complete	5	0.630	3,680	3
Average	3	0.789	3,743	3
Average	4	0.631	3,705	3
Average	5	0.616	3,702	3
Single	3	0.820	3,743	1
Single	4	0.789	3,743	1
Single	5	0.607	3,740	1

Interpretation: Complete, average and single linkage show high silhouette scores (0.61 to 0.82), but this is misleading. In these cases almost every customer falls into one huge

cluster (up to 3,743 of 3,749 customers) and the other clusters hold only a handful of customers. A useful segmentation cannot be built like this. Ward linkage gives more balanced groups, so it was chosen. With four clusters it has a silhouette score of 0.287 and cluster sizes of 2,499, 983, 229 and 38 customers.

4.3 DBSCAN Clustering

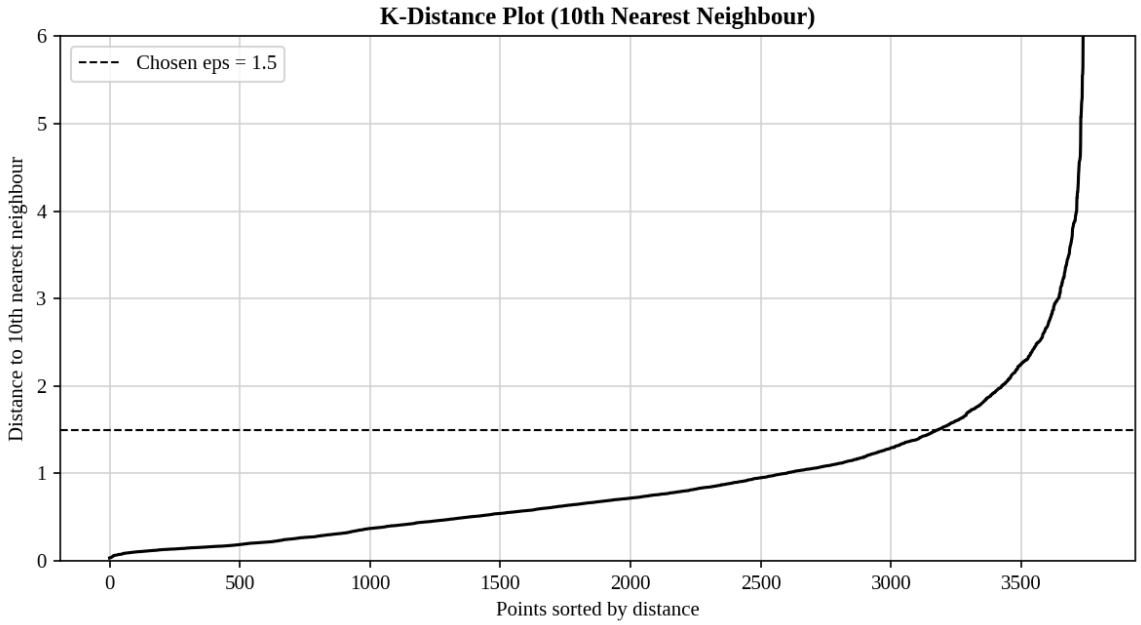


Figure 8: K-distance plot used to choose the DBSCAN eps value

Interpretation: The line rises slowly for most customers and then climbs quickly after a distance of about 1.3 to 1.9. This bend is a good region for eps, so eps = 1.5 was chosen (shown by the dashed line). The value of minPts was set to 10 after trying 5, 10 and 20.

Table 5: DBSCAN results for different eps and minPts values

eps	minPts	Clusters	Noise points	Silhouette
0.8	5	14	935	0.154
0.8	10	4	1,186	0.273
0.8	20	4	1,436	0.277
1.0	5	7	637	0.259
1.0	10	2	821	0.302
1.0	20	3	1,014	0.277
1.2	5	9	470	0.217
1.2	10	3	582	0.285
1.2	20	2	714	0.296
1.5	5	6	278	0.214
1.5	10	4	370	0.255
1.5	20	2	469	0.277
2.0	5	5	145	0.257
2.0	10	3	193	0.259
2.0	20	2	281	0.263

Interpretation: Small eps values mark many customers as noise (up to 1,436 customers), while large values merge everything into two or three clusters. The setting eps = 1.5 and minPts = 10 gives four clusters, marks 370 customers (about 10%) as noise and has a silhouette score of 0.255. The four clusters contain 2,291, 1,065, 13 and 10 customers, so DBSCAN effectively finds two large groups and two very small ones.

4.4 Visualising the Clusters

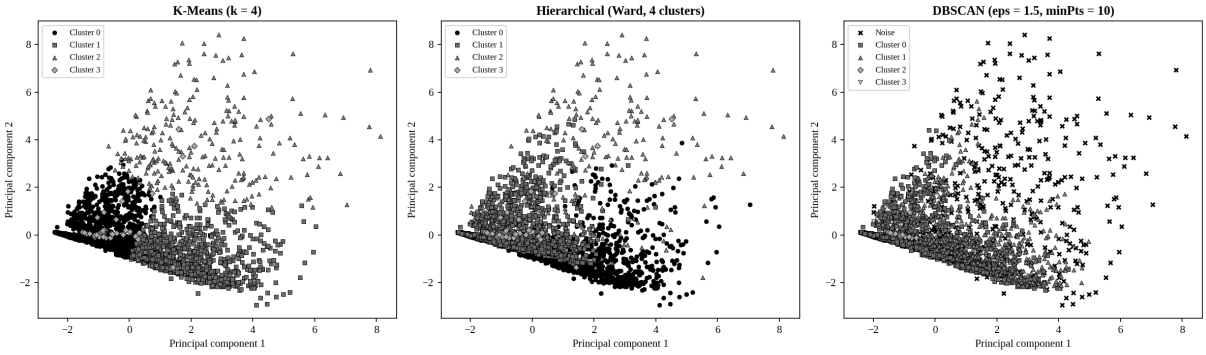


Figure 9: Clusters from the three algorithms shown on the first two principal components

Interpretation: The two principal components together explain about 45% of the total variation (28.8% and 16.6%), so the picture is only a partial view of the data. The clusters are not far apart and they overlap, which matches the modest silhouette scores. In the K-Means panel, the triangles (heavy flyers) spread towards the top of the plot, the circles (low-activity customers) sit on the left, and the squares (customers with high bonus miles) lie on the lower right. In the DBSCAN panel, the crosses are the noise points, and they are mostly the customers with the highest values spread far from the main crowd.

4.5 Cluster Profiles

Table 6: Average values of each variable for the K-Means clusters

Variable	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Customers	2,229	1,231	251	38
Balance	38,196	95,737	101,096	42,433
Qual miles	97	123	489	0
cc1 miles	1.1	3.6	2.0	1.1
cc2 miles	1.0	1.0	1.0	2.4
cc3 miles	1.0	1.0	1.0	1.0
Bonus miles	3,648	33,232	19,092	14,358
Bonus trans	6.1	17.4	16.1	14.9
Flight miles 12mo	102	154	2,262	285
Flight trans 12	0.4	0.5	6.4	1.1
Days since enroll	3,655	4,717	4,400	3,917
Award (share)	0.2	0.6	0.6	0.3

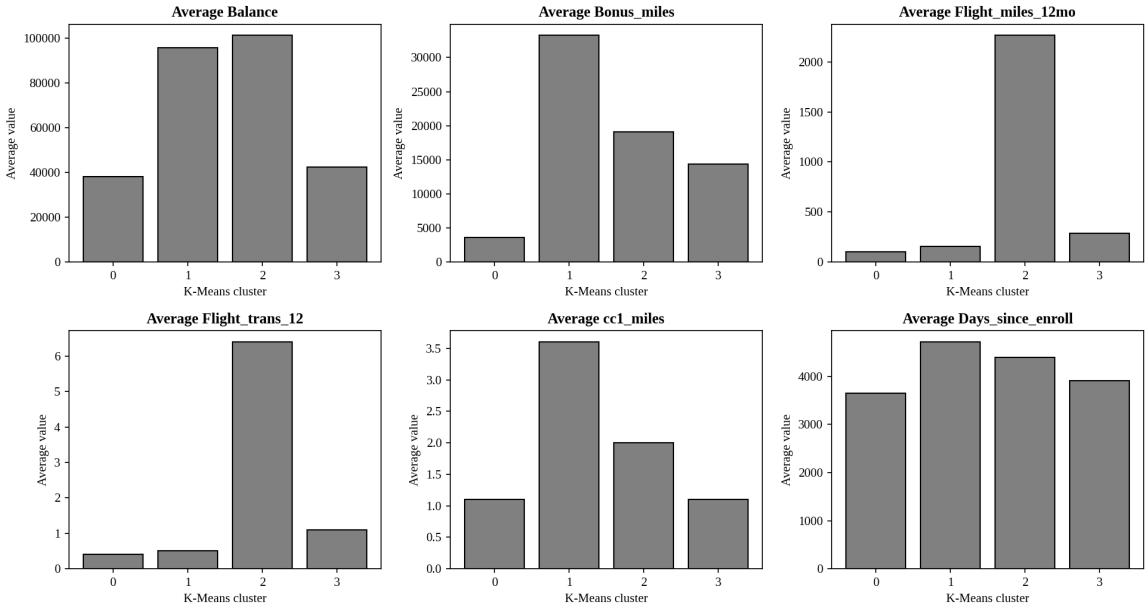


Figure 10: Average value of six key variables in each K-Means cluster

The K-Means clusters can be described as follows:

- **Cluster 0, Low-activity customers (2,229 customers, 59.5%):** This is the largest group. It has the lowest balance (about 38,200 miles), very few bonus miles (about 3,650), almost no flights (0.4 flight transactions) and the lowest card use (cc1 about 1.1). Only 20% have received an award.
- **Cluster 1, Credit card and bonus customers (1,231 customers, 32.8%):** These customers earn most of their miles from cards and bonus transactions. They have the highest bonus miles (about 33,200), the highest cc1 level (3.6), a high balance (about 95,700) and the longest membership (about 4,700 days). They fly very little (about 154 miles), and 60% have received an award.
- **Cluster 2, Frequent flyers (251 customers, 6.7%):** These customers fly far more than anyone else, with about 2,260 flight miles and 6.4 flight transactions. They also have the highest qualifying miles (about 489) and the highest balance (about 101,100).
- **Cluster 3, Rewards card customers (38 customers, 1.0%):** This small group is the only one that earns miles with the Rewards card (cc2 about 2.4). They have no qualifying miles and a low balance (about 42,400).

The other two algorithms give a similar picture. Ward hierarchical clustering finds a large low-activity group (2,499 customers), a card and bonus group (983 customers, cc1 about 3.8), a frequent flyer group (229 customers) and the same 38-customer Rewards card group, which has identical averages to K-Means Cluster 3. The fact that two different methods find the same four types of customers increases confidence in these segments. DBSCAN splits the main crowd almost completely by award status: its largest cluster

(2,291 customers) contains no award holders and its second cluster (1,065 customers) contains only award holders. DBSCAN treats the heaviest users (average balance of about 120,800 miles and 1,426 flight miles) as noise rather than as a segment of their own.

Table 7: Key averages for the hierarchical and DBSCAN clusters

Cluster	Customers	Balance	Bonus miles	Flight miles	cc1	cc2	Award
Hierarchical 0	983	97,439	37,157	221	3.8	1.0	0.4
Hierarchical 1	2,499	43,681	5,060	148	1.3	1.0	0.3
Hierarchical 2	229	102,284	20,360	1,740	2.1	1.0	0.6
Hierarchical 3	38	42,433	14,358	285	1.1	2.4	0.3
DBSCAN Noise	370	120,759	25,928	1,426	2.4	1.1	0.6
DBSCAN 0	2,291	53,029	9,046	96	1.7	1.0	0.0
DBSCAN 1	1,065	59,240	22,461	231	2.6	1.0	1.0
DBSCAN 2	13	34,806	8,390	19	1.0	2.0	0.0
DBSCAN 3	10	26,392	2,939	40	1.1	1.0	0.0

4.6 Comparing the Algorithms

Table 8: Comparison of the three clustering algorithms

Algorithm	Clusters	Noise points	Silhouette score
K-Means	4	0	0.319
Hierarchical (Ward)	4	0	0.287
DBSCAN	4	370	0.255

Interpretation: K-Means has the best silhouette score (0.319), followed by hierarchical clustering (0.287) and DBSCAN (0.255). All three scores are modest, which means the segments overlap to some extent. This is reasonable for customer behaviour data, where people do not fall into perfectly separate groups. K-Means also assigns every customer to a group and gives clusters that are easy to describe, so it is the most useful method for this problem.

5. Conclusion

Customers of East-West Airlines can be grouped into four meaningful segments: low-activity customers, credit card and bonus customers, frequent flyers, and a small group of Rewards card users. K-Means gave the best result with a silhouette score of 0.319, and the hierarchical method found almost the same segments. DBSCAN was less suitable because it marked about 10% of customers as noise, but it helped to show that award status clearly separates customers.

These segments can help the airline plan its offers:

- Low-activity customers form almost 60% of the base, so offers that encourage them to fly or use the credit card could bring the largest improvement.
- Credit card and bonus customers are loyal and have high balances, so they should be encouraged to use their miles through awards and partner offers.
- Frequent flyers are few but fly the most, so they deserve status benefits and special service.
- The Rewards card group is very small, so the airline can decide whether to promote that card more widely.

The main limitations are that the silhouette scores are only moderate, and that the 249 removed outliers are very active customers who may be the most valuable. They should be studied separately in future work.

Case Study 5: Wine Quality Assessment Using Physicochemical Properties

1. Problem Statement

A wine company currently decides the quality of its wine only by sensory assessment, which means tasting by experts. Management wants to know whether laboratory measurements of the wine can explain differences in quality and support a more data-driven quality check.

The aim of this case study is to find which physicochemical characteristics are most strongly related to wine quality, to build a multiple linear regression model that predicts the quality score from the 11 laboratory measurements, and to decide whether such a model could be used in a quality screening process.

2. Methodology

2.1 Dataset

The data comes from the Vinho Verde wines of Portugal (Cortez et al., 2009). There are two files, one for red wine with 1,599 samples and one for white wine with 4,898 samples. Both have the same 11 laboratory measurements: fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. The quality score is the median of at least three expert ratings on a scale from 0 to 10. The two files were joined into one dataset of 6,497 wines, and this combined data was used for the analysis. A separate comparison of red and white wine is given in Section 4.

There are no missing values. The dataset has 1,179 repeated rows. They were kept because different wines can genuinely have the same measurements and the same score, and the task did not ask for them to be removed.

2.2 Methods Used

- **Descriptive statistics:** mean, median, mode, standard deviation and skewness, plus the coefficient of variation to compare variability.
- **Pearson correlation:** to measure the strength and direction of straight-line relationships.
- **Graphs:** a histogram, a box plot and a correlation heatmap.
- **Multiple linear regression:** trained on 80% of the wines (5,197) and tested on the remaining 20% (1,300), using a fixed random state of 42 so that the split can be repeated.
- **Metrics:** MAE, MSE, RMSE and R-squared on the test set.
- **Standardised coefficients and VIF:** standardised coefficients compare the influence of predictors that are measured in different units, and the Variance Inflation Factor (VIF) checks multicollinearity.

3. Statistical Findings

3.1 Task 1: Statistical Analysis

Table 9: Descriptive statistics for alcohol, volatile acidity and residual sugar

Attribute	Mean	Median	Mode	Std. Dev.	Skewness
Alcohol	10.4918	10.30	9.50	1.1927	0.5657
Volatile acidity	0.3397	0.29	0.28	0.1646	1.4951
Residual sugar	5.4432	3.00	2.00	4.7578	1.4354

Interpretation: For all three attributes the mean is larger than the median, and the median is larger than the mode. This means the distributions are skewed to the right. Alcohol is only slightly skewed (0.57). Volatile acidity (1.50) and residual sugar (1.44) are strongly skewed, meaning most wines have low values but a few wines have very high values. For example, residual sugar has a median of 3.0 but a maximum of 65.8.

Table 10: Coefficient of variation

Attribute	Std. Dev.	Mean	Coefficient of variation (%)
Alcohol	1.1927	10.4918	11.37
Volatile acidity	0.1646	0.3397	48.47
Residual sugar	4.7578	5.4432	87.41

Interpretation: The three attributes are measured in different units, so their standard deviations cannot be compared directly. The coefficient of variation (standard deviation divided by the mean, shown as a percentage) removes the units. It is 11.37% for alcohol, 48.47% for volatile acidity and 87.41% for residual sugar. **Residual sugar has the greatest relative variability.** Its values spread widely compared with its average, while alcohol is the most consistent.

Alcohol and quality: Pearson's correlation coefficient between alcohol and quality is r = 0.4443, with a p-value far below 0.001. The relationship is positive and moderate: wines with higher alcohol content tend to receive higher quality scores. Squaring r gives about 0.20, so alcohol alone accounts for roughly 20% of the differences in quality. Correlation does not prove that alcohol causes better quality.

3.2 Task 2: Exploratory Data Analysis

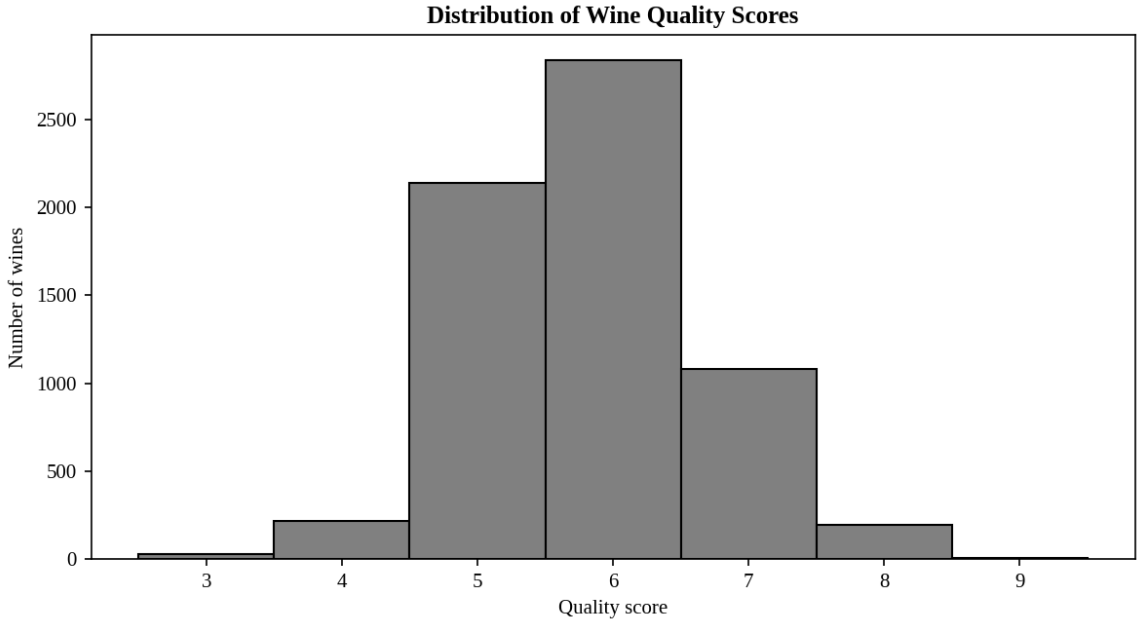


Figure 11: Distribution of wine quality scores

Interpretation: The scores range from 3 to 9, and the histogram is roughly bell-shaped with a slight tail to the right (skewness 0.19). The most common score is 6 (2,836 wines, 43.7%), followed by 5 (2,138 wines, 32.9%) and 7 (1,079 wines, 16.6%). The average is 5.82. Very poor wines (score 3, only 30) and excellent wines (score 9, only 5) are extremely rare. This imbalance will make it hard for any model to predict the extreme scores.

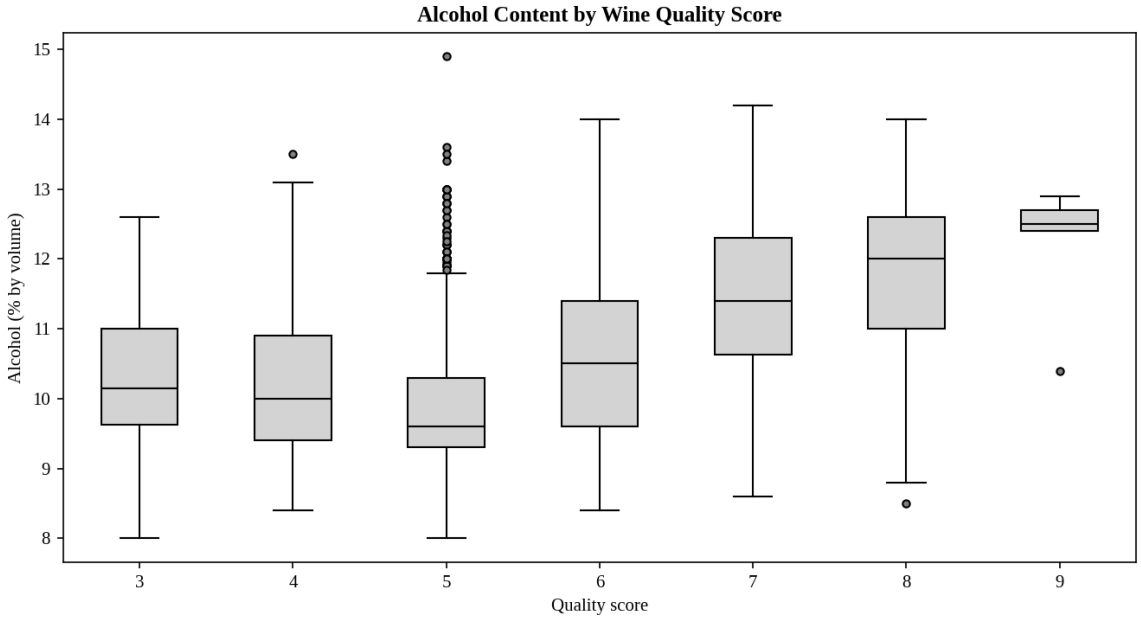


Figure 12: Alcohol content grouped by wine quality score

Interpretation: The median alcohol is 10.15 for score 3, 10.0 for score 4 and 9.6 for score 5, then it rises to 10.5 for score 6, 11.4 for score 7, 12.0 for score 8 and 12.5 for score 9. So wines with scores of 6 and above clearly have more alcohol than average-quality wines. Outliers (the dots) appear in several groups. Quality 5 has many high-alcohol outliers, reaching about 14.9, and quality 4, 8 and 9 each have a few. The quality 9 box is based on only five wines, so it should be read with care.

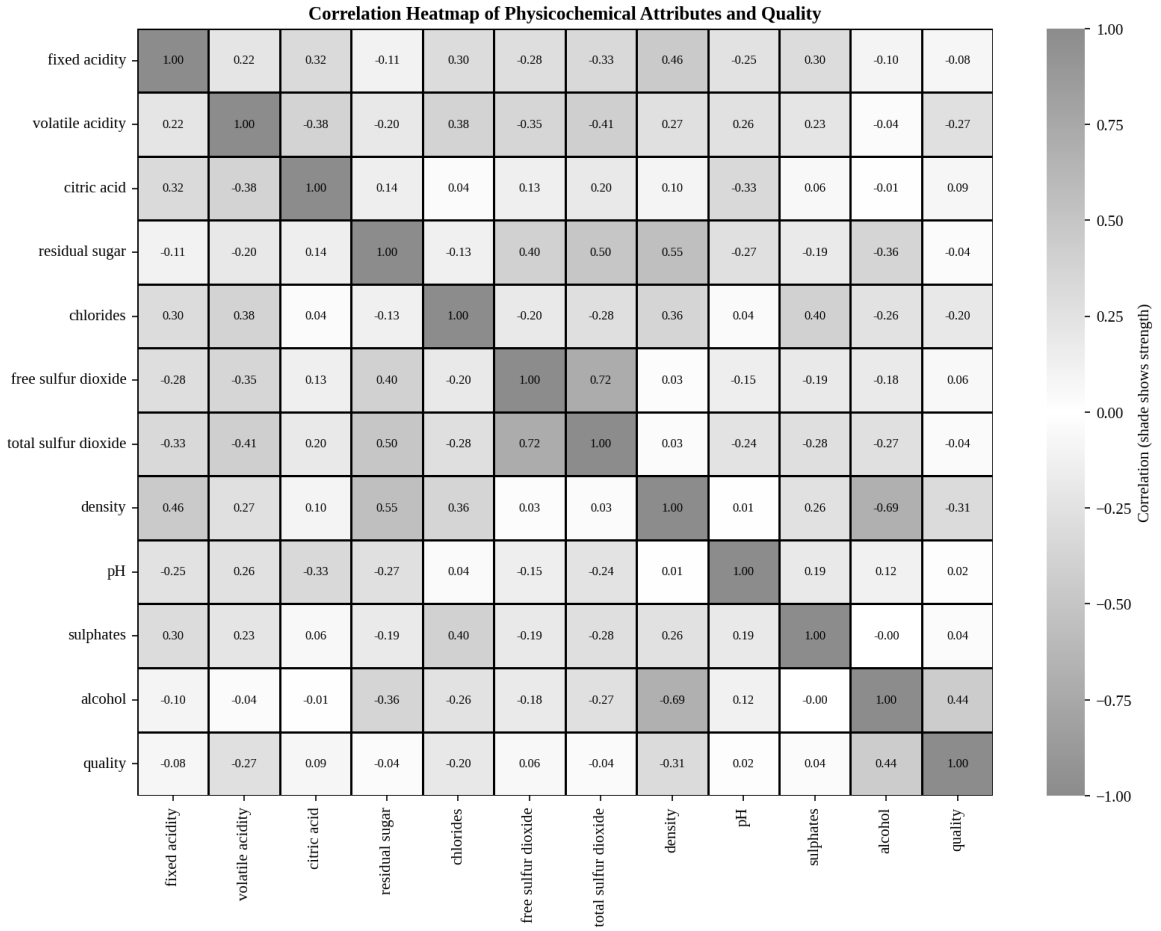


Figure 13: Correlation heatmap of physicochemical attributes and quality

Interpretation: Darker boxes show stronger relationships and the number gives the sign. The strongest relationships are between free and total sulfur dioxide (0.72), alcohol and density (-0.69), and residual sugar and density (0.55). Total sulfur dioxide is also related to residual sugar (0.50). These strong links among the attributes warn us that multicollinearity may be present in the regression model. With quality, the strongest links are alcohol (0.44), density (-0.31), volatile acidity (-0.27) and chlorides (-0.20).

Table 11: Pearson correlation of each attribute with quality, ordered by absolute value

Attribute	Correlation with quality	Direction
Alcohol	0.4443	Positive
Density	-0.3059	Negative
Volatile acidity	-0.2657	Negative
Chlorides	-0.2007	Negative
Citric acid	0.0855	Positive
Fixed acidity	-0.0767	Negative
Free sulfur dioxide	0.0555	Positive
Total sulfur dioxide	-0.0414	Negative
Sulphates	0.0385	Positive
Residual sugar	-0.0370	Negative
Ph	0.0195	Positive

The two attributes most strongly associated with quality are:

- **Alcohol (r = +0.444), a positive association:** wines with more alcohol tend to be rated higher.
- **Density (r = -0.306), a negative association:** denser wines tend to be rated lower. Density is closely tied to alcohol (r = -0.69), because alcohol is lighter than water and lowers the density of a wine. Part of the link between density and quality is therefore the same effect as alcohol.

All other attributes have an absolute correlation below 0.27 with quality.

4. Model Evaluation

4.1 Task 3: Multiple Linear Regression

A multiple linear regression model was trained on the 11 attributes using 5,197 wines, and its predictions were checked on the 1,300 wines that were not used for training. The table below shows the test-set metrics.

Table 12: Test-set performance of the regression model

Metric	Value	Meaning
MAE	0.5659	On average, a prediction is wrong by about 0.57 quality points
MSE	0.5467	Average of squared errors; large mistakes count more
RMSE	0.7394	Typical error size in quality points, about 0.74
R-squared	0.2598	The model explains about 26% of the differences in quality

Interpretation: The average error of 0.57 may look small, but the scores only run from 3 to 9 and most wines score 5 or 6, so an error of half a point can change a good wine into an average one. About 53.9% of the predictions are within 0.5 points of the true score and 84.2% are within 1 point, so roughly one wine in six is wrong by more than a full point. An R-squared of 0.26 means that about 74% of the variation in quality is not explained by the laboratory measurements in this model.

Influential predictors

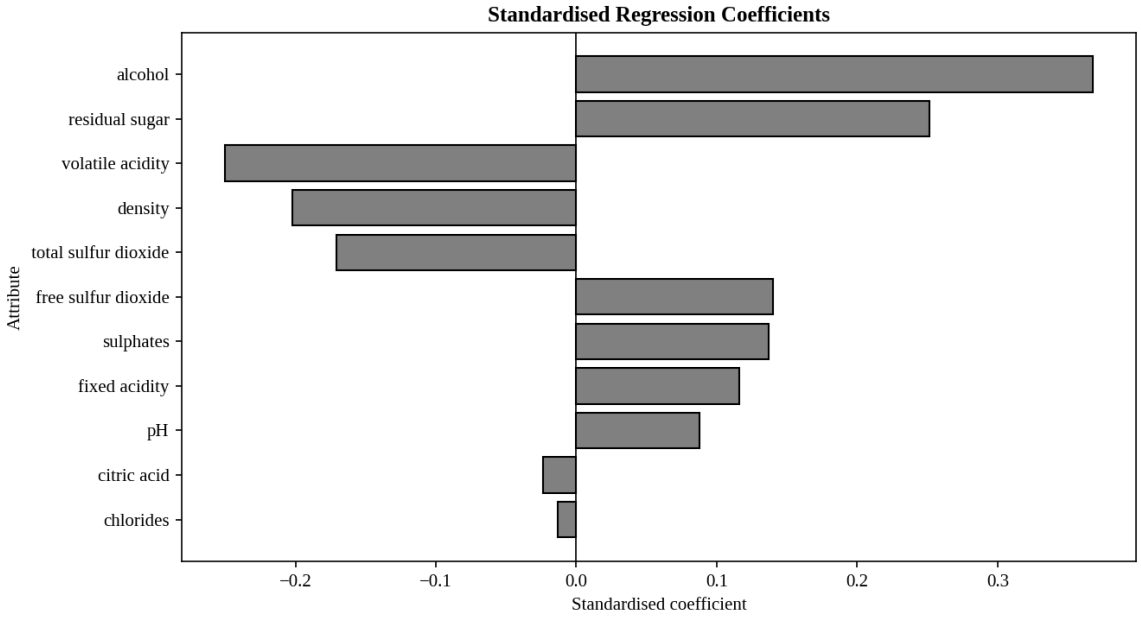


Figure 14: Standardised regression coefficients of the 11 attributes

Table 13: Standardised coefficients and VIF values, ordered by absolute coefficient

Attribute	Standardised coefficient	VIF
Alcohol	+0.3678	4.62
Residual sugar	+0.2514	7.20
Volatile acidity	-0.2499	1.94
Density	-0.2023	15.51
Total sulfur dioxide	-0.1706	2.96
Free sulfur dioxide	+0.1398	2.16
Sulphates	+0.1371	1.54
Fixed acidity	+0.1161	4.74
Ph	+0.0879	2.47
Citric acid	-0.0237	1.61
Chlorides	-0.0132	1.65

Multicollinearity first: A VIF above 5 is a sign of concern and a VIF above 10 is serious. Density has a VIF of 15.51, which is very high. Residual sugar (7.20), fixed acidity (4.74) and alcohol (4.62) are also moderately high, while all other attributes are below 3. This means density, sugar and alcohol carry overlapping information, so their individual coefficients can be unstable and should be read with care.

Most influential predictors: By absolute standardised coefficient, the two most influential predictors are alcohol (+0.368) and residual sugar (+0.251). Volatile acidity (-0.250) is almost tied with residual sugar, so the order of second and third place is practically equal.

- **Alcohol** has the same positive sign as its simple correlation with quality (+0.44), and a VIF of 4.62 is only moderate. Its effect can be trusted: with the other measurements held fixed, higher alcohol goes with higher quality.
- **Residual sugar** needs caution. On its own it has almost no relationship with quality (r = -0.04), yet its coefficient in the model is positive and large. This change of sign happens because sugar is strongly related to density (VIF 15.51 for density), and the model uses sugar to adjust for density. It should not be understood as 'more sugar makes better wine'.
- **Volatile acidity** has a low VIF (1.94), a negative sign that agrees with its simple correlation, and a standardised coefficient almost as large as that of residual sugar. Its effect is therefore easier to trust.

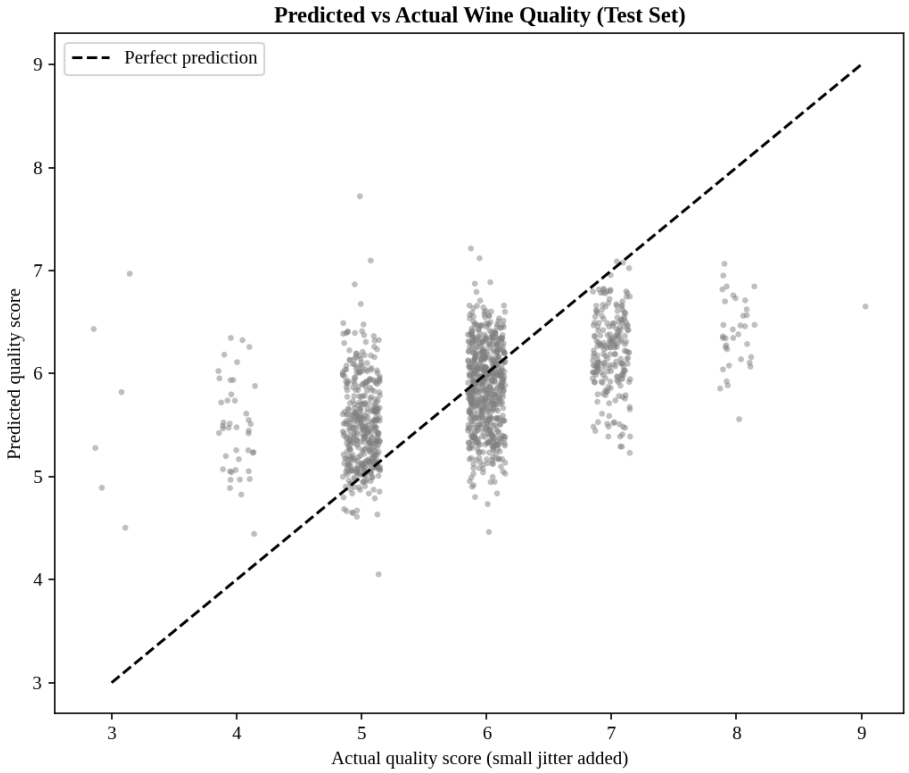


Figure 15: Predicted against actual quality scores on the test set

Interpretation and limitations: The dashed line shows perfect predictions. The points form vertical stripes because actual scores are whole numbers, and they follow the line only loosely. The predictions stay between 4.05 and 7.72, so the model never predicts the lowest or the highest scores. Wines with scores of 3 and 4 are predicted too high and wines with scores of 8 and 9 are predicted too low. The model pulls everything towards the average. The main limitations are:

- The model can only draw straight-line relationships, while real wine quality may depend on more complex combinations of measurements.
- Multicollinearity makes some coefficients, especially for density and residual sugar, unreliable to interpret.
- The quality scores are subjective opinions and are crowded between 5 and 7, so the model has very few examples of poor and excellent wines.
- Red and white wines were combined, although they may behave differently.

Red wine and white wine separately

Table 14: Test-set results for models trained on each dataset

Dataset	Rows	MAE	RMSE	R-squared
Red wine	1,599	0.504	0.625	0.403
White wine	4,898	0.586	0.754	0.265
Combined	6,497	0.566	0.739	0.260

Interpretation: The model for red wine alone performs better (R-squared 0.403) than the models for white wine (0.265) or the combined data (0.260). This suggests that red and white wines are influenced differently by the measurements. Even so, the best model explains only about 40% of the variation in quality.

5. Conclusion and Business Recommendations

5.1 Task 4: Business Decision-Making

a. Measurements that deserve attention during screening

- **Alcohol:** It has the strongest correlation with quality (+0.44) and the largest standardised coefficient (+0.368). Wines with quality scores of 7 and above have a median alcohol of 11.4% or more, compared with 9.6% for score 5. It is also a simple and cheap measurement to take.
- **Volatile acidity:** It has a clear negative link with quality (r = -0.27, standardised coefficient -0.250). Unlike density and residual sugar, its VIF is low (1.94), so its effect is stable and easy to explain. High volatile acidity is also known to give wine an unpleasant, vinegar-like smell.

Density was the second strongest in correlation, but it mostly repeats the information in alcohol and sugar (VIF 15.51), so it adds little as a separate screening measurement.

b. Can the model replace sensory evaluation? No. The regression model is not reliable enough to replace sensory evaluation. On the test set, R-squared is only 0.26, so about 74% of the variation in quality is unexplained. The RMSE of 0.74 and MAE of 0.57 are large compared with the narrow range in which most scores fall, and only about 54% of the predictions are within half a point. The model also cannot predict the best and the worst wines (its predictions lie between 4.05 and 7.72), and these are the wines the company cares about most. The model can still be useful as a support tool. For example, wines with low alcohol or high volatile acidity could be flagged for a closer tasting, while the final judgement stays with the expert panel.

c. Two limitations of the dataset and the approach

- **Subjective and unbalanced quality scores:** The quality score is based on the opinion of tasters, and about 93% of the wines have scores of 5, 6 or 7. There are very few poor or excellent wines to learn from, so the model cannot predict them well.
- **Limited data and a simple model:** All wines come from one region (Vinho Verde in Portugal) and no information about grape type, brand or price is included, so the results may not apply to other wines. The linear model also assumes straight-line effects and is affected by multicollinearity, which limits how well it can explain quality.

Overall conclusion: Alcohol and volatile acidity are the most useful laboratory measurements for judging wine quality. A laboratory-based screening process is a good addition to the company's quality control, but it should support sensory evaluation and not replace it.

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